MySQL & Power BI Portfolio Roadmap (UC California & Coursera)

Data Analysis: 3 Phases Portfolio Roadmap

A strategically designed learning path that builds three portfolio projects showcasing my SQL skills. Each project simulates real business scenarios relevant to data analyst roles.

My portfolio consists of:

- **Phase 1 Project: E-Commerce Database Design & SQL Fundamentals**
- **Phase 2 Project: Marketing Campaign Analytics Dashboard**
- **Phase 3 Project: Customer Churn and Loyalty Segmentation**

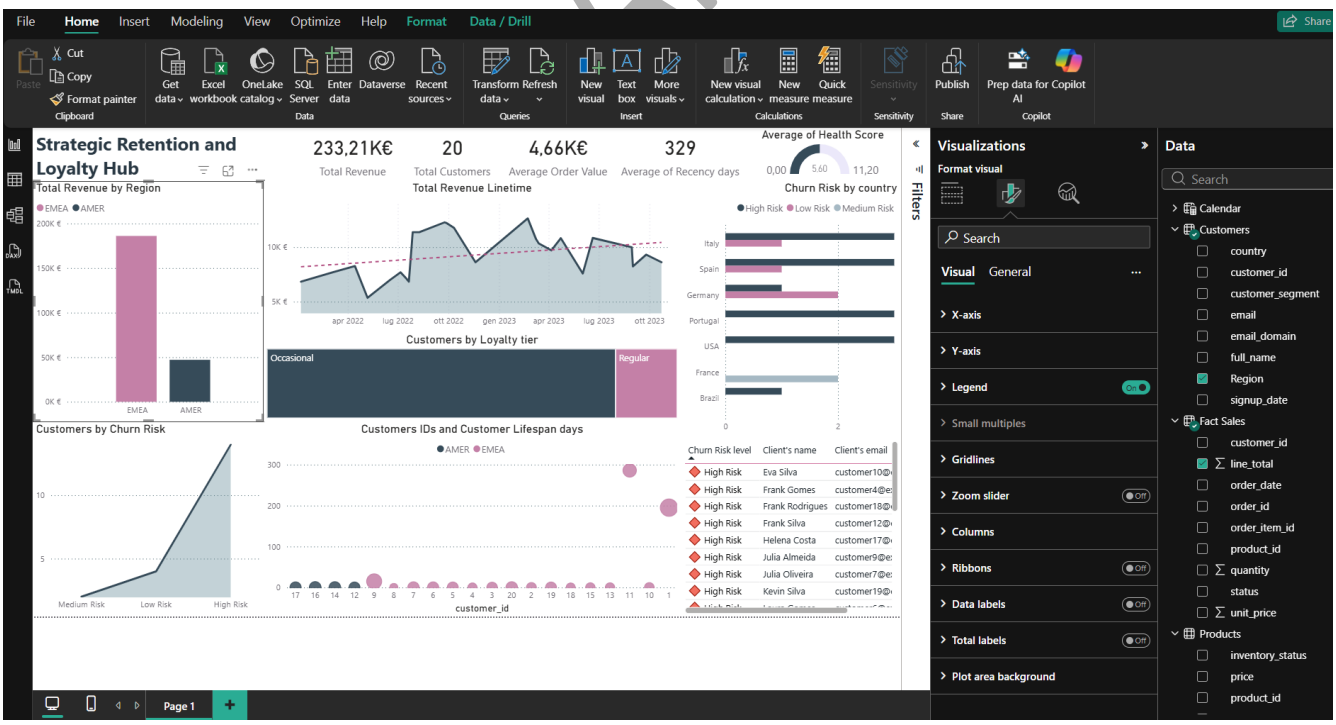
Lessons and projects are based on course of Coursera: SQL for Data Science of University of California, Davis:
https://www.coursera.org/learn/sql-for-data-science

Arroyo Tanta

Presentation:

With this documentation I welcome the reader and present my skills in SQL language, especially using MySQL for data analysis and Business Intelligence. Each phase reflects new challenges using datasets created with Perplexity and following the lessons of the course "SQL for Data Science of University of California". The complexity of this program increases progressively without detracting from the previous steps, culminating in a project using Power BI that integrates data modeling, DAX functions, cleaning in Power Query and the creation of dashboards with interactive charts.

I hope this content is of interest to the reader and for security reasons and limited resources, this document includes only screenshots and a download link for the viewing mode; Full access to the documentation will be available only by direct contact with me.



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SQL content:

Phase 1:

Learning Objectives of Course

- Master SELECT, WHERE, ORDER BY for data retrieval
- Understand relational database design (primary/foreign keys, normalization)
- Create production-ready table schemas
- Validate data integrity

Phase 2:

Learning Objectives

- Combine multiple tables using INNER, LEFT, RIGHT joins
- Aggregate data with COUNT, SUM, AVG, MIN, MAX
- Group and filter aggregated data (GROUP BY, HAVING)
- Use subqueries for complex filtering
- Understand join performance implications

Phase 3:

Learning Objectives

- Transform text data with CONCAT, SUBSTRING, TRIM, UPPER/LOWER
- Handle dates and temporal analysis
- Create conditional features with CASE statements
- Build reusable views
- Validate data quality

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Phase 1: SQL Fundamentals + Database Design Project

Phase 1 Portfolio Project: E-Commerce Database Design & Exploratory Analysis

Business Scenario: Hired as a data analyst at a mid-size e-commerce startup, my first task is to validate the database structure and run exploratory queries to better understand the business.
For this project, I used Laragon and HeidiSQL to create the data tables, upload the datasets, and validate the business queries. I also generated a synthetic dataset with Perplexity, and I used Power BI (data modeling and Power Query) to build the dashboard and the "Business Vision" section.

Learning Objectives of Course

- Master SELECT, WHERE, ORDER BY for data retrieval
- Understand relational database design (primary/foreign keys, normalization)
- Create production-ready table schemas
- Validate data integrity

Topics (Lessons 1-9)

Days 1-2: SELECT & Data Retrieval (Lessons 1, 4)

- SELECT single/multiple columns, SELECT *
- LIMIT and OFFSET for pagination
- Comments for code documentation

Goal: Retrieve data efficiently without loading everything into memory

Days 2-3: Table Design & Data Loading (Lesson 2)

- Data types: INT, VARCHAR, DECIMAL, DATE, BOOLEAN, TEXT
- Constraints: PRIMARY KEY, NOT NULL, UNIQUE, DEFAULT
- AUTO_INCREMENT for surrogate keys
- INSERT strategies (bulk loading vs single records)

Goal: Design normalized schemas that prevent data anomalies

Day 3: Temporary Tables & Intermediate Storage (Lesson 3)

- CREATE TEMPORARY TABLE for multi-step analyses
- When to use vs when to optimize

Goal: Handle complex transformations step-by-step

Days 4-5: Filtering Basics (Lesson 5)

- WHERE conditions: =, !=, <, >, <=, >=
- String filtering and LIKE wildcards
- BETWEEN for ranges
- IS NULL / IS NOT NULL

Goal: Extract exact subsets without over-fetching data

Days 5-6: Advanced Filtering (Lesson 6)

- IN operator (multiple values)
- AND / OR logical operators
- NOT for exclusions
- Operator precedence and parentheses

Goal: Write complex WHERE conditions that mirror business rules

Days 6-7: Patterns & Wildcards (Lesson 7)

- LIKE with % and _ wildcards
- REGEXP for pattern matching

Goal: Search unstructured data reliably

Day 7: Sorting & Math (Lessons 8-9)

- ORDER BY single/multiple columns
- ASC/DESC for direction
- Mathematical operators: +, -, *, /, parentheses
- Calculated columns with aliases

Goal: Prepare data for analysis and storytelling

Phase 1 Portfolio Project:

1. Database Schema Design (DDL Script):

Create normalized schema for an e-commerce company with:

```
-- Tables (normalized 3NF design):
- customers (customer_id, first_name, last_name, email, country, signup_date, lifetime_spend)
- products (product_id, name, category, price, stock_quantity, supplier_id)
- suppliers (supplier_id, company_name, contact_email, country)
- orders (order_id, customer_id, order_date, total_amount, status)
- order_items (order_item_id, order_id, product_id, quantity, unit_price)
- categories (category_id, category_name, parent_category_id) [optional hierarchy]
```

- Proper primary/foreign key relationships
- Appropriate data types (no VARCHAR for dates)
- NOT NULL constraints where needed
- DECIMAL(10,2) for money columns

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- Clear, professional table naming

2. Sample Data Population (INSERT Script)

Insert 50 realistic records across all tables showing:

- Diverse customers (different countries, signup dates)
- Multiple product categories
- Realistic order patterns (including cancelled, pending, completed orders)
- Different price points and inventory levels

3. Exploratory Data Analysis Queries

GROUP 1: BASIC RETRIEVAL

Query 1
Name: Oldest 10 Customers
Business Context: Identify loyal, early customers
Expected Output: 10 rows, oldest customers first

Query 2
Name: Premium Products (>$100)
Business Context: Find premium products for high-value inventory analysis and premium customer targeting
Expected Output: All products priced over $100, ordered from highest to lowest price

Query 3
Name: Orders from Last 30 Days
Business Context: Monitor recent order activity
Expected Output: Orders placed in last 30 days, most recent first

Query 4
Name: 5 Newest Orders
Business Context: Quick view of latest transactions
Expected Output: Exactly 5 rows, most recent orders

Query 5
Name: Brazil Customers
Business Context: Geographic segmentation for targeted campaigns
Expected Output: All customers from Brazil, ordered alphabetically by last name

GROUP 2: FILTERING & DATA UNDERSTANDING

Query 6
Name: Cancelled Orders
Business Context: Data quality - ensure cancelled orders are properly recorded
Expected Output: Only cancelled orders, newest first

Query 7
Name: Out of Stock Products
Business Context: Inventory management - identify products to restock
Expected Output: All products with zero inventory, alphabetically ordered

Query 8
Name: Customers with Missing Country (Data Quality Check)
Business Context: Data quality - identify records needing correction
Expected Output: All customers missing country information

Query 9
Name: Orders in Mid-Range ($100-$500)
Business Context: Segment orders by value range
Expected Output: Orders in price range, highest value first

Query 10
Name: Products Outside Price Range ($50-$200)
Business Context: Identify price outliers (very cheap or very expensive)
Expected Output: Products outside mid-range, lowest price first

GROUP 3: ADVANCED FILTERING & SORTING

Name: Customers from Multiple Countries
Business Context: Multi-country campaign targeting
Expected Output: Customers from 4 specific countries, grouped by country then alphabetically

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Query 12
Name: Clean Orders (Exclude Problematic Status)
Business Context: Filter to clean transaction data (exclude problematic statuses)
Expected Output: Only completed/valid orders, most recent first

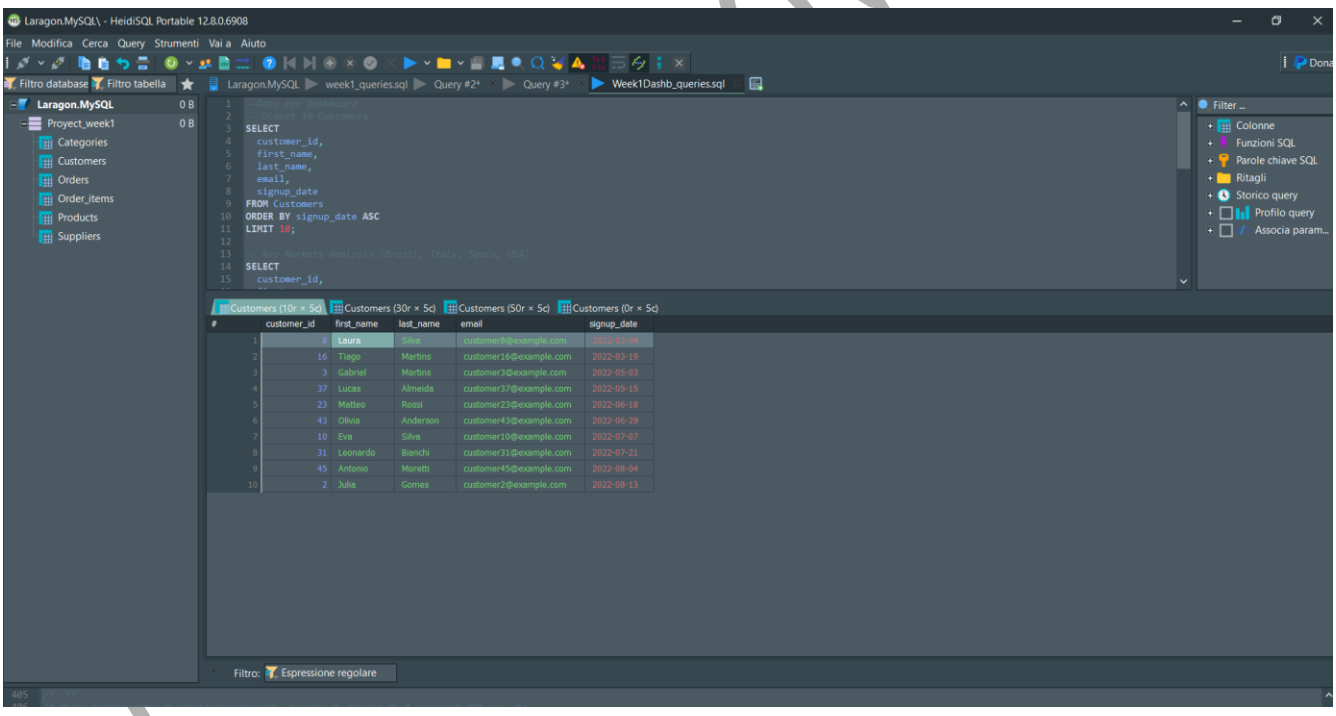
Query 13
Name: Email Customers
Business Context: Email provider analysis for communication strategy
Expected Output: Customers with "@example.com" email addresses, ordered by email

Query 14
Name: Electronics or Home Products
Business Context: Filtering electronics and home appliances for promotion
Expected Output: Products with "Laptop", "Smartphone", "Blender", "Toaster" ordered by implicit category and price.

Query 15
Name: Order Analysis with Math Calculation
Business Context: Financial analysis at the order level (15% discount and simulated 21% tax)
Expected Output: Top 10 orders with calculated fields, highest total first.

4. Business Vision:

By selecting queries 1, 11, 13, and 8, we obtain key information for decision-making. Specifically, we can identify the 10 most loyal customers and analyze business performance across four different countries, as well as determine how many customers come from each one.
We can also access the quality of our data by detecting customers with missing information in critical fields such as country (with result of 0).
Finally, if we want to analyze our customers' email addresses, we can classify them by domain to identify potentially fake accounts created with temporary email services. This allows us to focus our efforts on customers with verifiable addresses and reduce the costs associated with CRM tools.
Taken together, these queries strengthen both business analysis and database integrity, enabling more precise and efficient decision-making.



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5. Documentation (README.md)

README.md - (overview & instructions,how to use)
Carpet "SQL"

- week1_schema (it contains all the schema for the tables and its connections)
- week1_dataLoadMySQL (it contains all the code to insert the data manually or via code for MySQL)
- week1BusQueries (it contains all the queries to resolve the business needs)
- week1DashQueries (it contains all the queries for PowerBI or Excel)

Carpet "records" (it contains the data for business needs in csv files)
Carpet "dashResults" (it contains the results of week1DashQueries and the Power BI dashboard).

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Phase 2: Data Integration & BI Analytics Project

Phase 2 Portfolio Project: Marketing Analytics Dashboard (SQL + Insights)

Business Scenario: A marketing director needs to understand campaign performance across channels to allocate the next budget. I'll build a BI dashboard using SQL code and Power BI to answers strategic questions.

Learning Objectives

- Combine multiple tables using INNER, LEFT, RIGHT joins
- Aggregate data with COUNT, SUM, AVG, MIN, MAX
- Group and filter aggregated data (GROUP BY, HAVING)
- Use subqueries for complex filtering
- Understand join performance implications

Topics (Lessons 10-19)

Days 1-2: Aggregate Functions (Lesson 10)

- COUNT(*) vs COUNT(column) vs COUNT(DISTINCT)
- AVG(), MIN(), MAX(), SUM()
- COALESCE() for NULL handling in aggregates

Goal: Summarize data for executive-level insights

Days 2-3: Grouping & Filtering Groups (Lessons 11-12)

- GROUP BY single and multiple columns
- HAVING clause (filter on aggregates, not rows)
- WHERE vs HAVING distinction

Goal: Create segment-level analyses (by region, category, time period)

Days 3-4: Subqueries (Lesson 13)

- Subqueries in WHERE clause
- IN, EXISTS operators with subqueries
- Scalar subqueries vs set-returning subqueries
- When to use subqueries vs JOINs

Goal: Handle complex filtering logic elegantly

Days 4-5: INNER JOINS (Lessons 14, 16)

- INNER JOIN syntax and behavior (intersection only)
- Joining 2, 3, 4+ tables
- Identifying matching vs non-matching records

Goal: Combine related data from multiple sources

Days 5-6: Advanced Joins (Lessons 15, 17-18)

- CROSS JOIN for combinations and testing
- Self joins for hierarchical data
- LEFT JOIN (keep all from left table)
- RIGHT JOIN (keep all from right table)
- Full outer joins using UNION

Goal: Handle missing or optional relationships

Day 7: Set Operations (Lesson 19)

- UNION (vertical combining with deduplication)
- UNION ALL (faster, keeps duplicates)

Goal: Append datasets from different sources

Phase 2 Portfolio Project:

1. Sample Database (From Your Document)

Use the Campaigns + Performance structure from your SQL document:

```
Tables:
- campaigns (campaign_id, client_name, platform, status, daily_budget, objective, start_date)
- performance_logs (log_id, campaign_id, date, clicks, impressions, conversions, spend)
- clients (client_id, client_name, industry, total_budget, contract_end_date).
```

2. Core Analytics Queries

Platform Analysis:

1. Revenue by platform (sum spend, clicks, conversions per platform)
2. Average CPC (cost per click) by platform - which is most efficient?

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3. Conversion rate by platform - which converts best?
4. Platform comparison with ranking (using ORDER BY)

Client Performance:
5. Top 10 clients by spend and ROI (conversions / spend)
6. Client segmentation: which spend >$2000 total?
7. Underperforming clients (low conversion rate)
8. Client concentration risk (top 3 clients = % of total spend)

Campaign Diagnostics:
9. Active campaigns with no conversions (identify problem campaigns)
10. Campaigns with budget exhaustion >90% (scaling opportunities)
11.Campaign age analysis (new vs mature, which performs better?)
12. Seasonal trends (if data spans multiple months)

Advanced Aggregations:
13. Conditional aggregation: budget allocation by status (active/paused/completed)
14. Multi-dimensional: spend by platform AND client AND status (3-way GROUP BY)

3.Complex JOIN Query with Executive Dashboard

Generating an "Executive Dashboard" to view, for each client and platform, the total spend, the total number of conversions, the average daily spend and the number of active campaigns, ordering the results to highlight the platforms with the highest total spend.
This query demonstrates:

- Multi-table joins
- Aggregation with GROUP BY
- Mathematical operations
- Sorting for insights
- Similar Executive Request

4. Data Profiling & Quality Checks

1. NULL value counts by column
2. Date range validation (are all dates within expected window?)
3. Join accuracy test (how many campaigns have 0 performance records?

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5. Business Vision:

To address the objective of our project, we have selected four key metrics that allow us to obtain an overall view of the performance of our Marketing Agency. First, we analyze the Total Spend, meaning how much we have spent in total, and with more precise data we could calculate the annual spend and compare it with the previous year. We also consider the total number of conversions and the average per campaign, which allows us to evaluate the combined performance of our four platforms. Finally, we analyze the total volume of clicks to understand to what extent these campaigns have been productive and able to capture the customer's attention and convert it into a conversion.

Since our final purpose is to guide the Agency's management on where to allocate budget in the next period, it is essential to understand the average cost per platform and how these behave when segmented by industry. This provides us with a broader context to decide whether it is appropriate to prioritize certain clients according to their economic activity or if we want to plan a new campaign while considering the historical behavior of each activity or type of business.

After that, we analyze the relationship between conversions, clicks, and the budget allocated per campaign, which helps us anticipate and guide possible scenarios for future campaigns. Finally, we review the history of campaigns without conversions, which allows us to foresee risks and reserve a buffer budget if this type of result has occurred repeatedly in the past.

6. Documentation (README.md)

README.md - (overview & instructions,how to use)

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Phase 3: Data Transformation & Feature Engineering Project

Phase 3 Portfolio Project: Customer Churn and Loyalty Segmentation

Business Scenario: An e-commerce company aims to segment its customers for targeted marketing campaigns. To achieve this, I will develop a customer intelligence system using SQL-based feature engineering. This system will culminate in a comprehensive Customer Churn and Loyalty Segmentation analysis.

Learning Objectives

- Transform text data with CONCAT, SUBSTRING, TRIM, UPPER/LOWER
- Handle dates and temporal analysis
- Create conditional features with CASE statements
- Build reusable views
- Validate data quality
- Prepare data for machine learning

Topics (Lessons 20-26)

Days 1-2: String Manipulation (Lesson 20

- CONCAT() to combine fields
- TRIM(), LTRIM(), RTRIM() to clean whitespace
- SUBSTRING() to extract parts
- UPPER(), LOWER() for standardization

Goal: Standardize messy real-world data

Days 2-3: Date Functions & Temporal Analysis (Lessons 21-22)

- DATE() to strip time from timestamp
- YEAR(), MONTH(), DAY() for extraction
- DATE_FORMAT() for custom formatting
- DATEDIFF(), DATE_ADD(), DATE_SUB() for calculations
- TIMESTAMPDIFF() for age calculations

Goal: Extract time-based features for analysis

Days 3-4: CASE Statements (Lessons 23-24)

- Simple CASE for binary flags (1/0)
- Searched CASE for complex conditions
- Data binning (Small/Medium/Large)
- Conditional aggregation (CASE inside SUM/COUNT)

Goal: Engineer features for segmentation and ML

Days 4-5: Views & Encapsulation (Lesson 25)

- CREATE VIEW for reusable queries
- Views as data abstraction layer
- CREATE OR REPLACE VIEW for updates

Goal: Build consistent data marts

Days 5-6: Data Profiling & Governance (Lesson 26)

- NULL counts and distributions
- DISTINCT value counts
- Data type validation
- Cardinality analysis

Goal: Understand data quality before analysis

Phase 3 Portfolio Project:

1. Database Setup

Tables:
- Orders: order_id, customer_id, order_date, total_amount, status
- Order_items: order_item_id, order_id, product_id, quantity, unit_price.
- Products: product_id, name, price, stock_quantity.
- Customers: customer_id, first_name, last_name, email, country, signup_date, lifetime_spend,average_purchase_value ,purchase_frequency, average_customer_lifespan_days, customer_lifetime_value.

2. Feature Engineering Queries

Text Standardization:

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1. Standardize customer names: trim whitespace, consistent casing
2. Extract domain from email addresses
3. Create customer segments based on email patterns (B2B vs B2C)

Date Features:
4. Customer age (days since signup)
5. Days since last order (recency)
6. Identify "new customers" (signup < 30 days ago)
7. Identify "dormant customers" (no order > 6 months)

Aggregated Features:
8. Customer lifetime value (sum of all order totals)
9. Order frequency (count of orders per customer)
10. Average order value per customer
11. Time between orders (for repeat customers)

CASE-Based Segmentation:
12. Budget tier: Small/Medium/Large based on spend
13. Activity status: Active/At-Risk/Dormant/New
14. VIP flag: Customers in top 20% by spend
15. Regional segments: based on country/region

3. Customer Churn and Loyalty Segmentation

Generating a SQL query to generate a report for each customer that includes:

- Customer ID, name, and email
- The date of their first order and last order
- The total number of days between their first and last order (customer activity span)
- The average number of days between their orders (for customers with more than one order)
- A churn risk segment using CASE
- A loyalty segment using CASE:

4.Reusable Views

Generating a SQL query to creating reusable views (Encapsulating all customer-level features) to answer:

- RFM segmentation for marketing team and updates daily/weekly as new orders arrive.???
- Composite score combining recency, frequency, monetary, used for churn prediction and retention prioritization

5. Data Quality Validation

- Ensure feature calculations are correct:
    - Verify no negative in Customer_health_score.
    - Check for NULL segments (missing data handling)
    - Validate date calculations (no future dates)
    - Spot-check: manually verify 5 Customer_health_score.

6. Marketing Application Queries

- Which customers are at risk of churning?
- Target high-value but inactive customers
- Growth opportunity: convert one-time buyers to repeat

7. Business Vision

The activity shows a clear concentration of revenue in the EMEA region, while AMER remains a secondary market. This distribution indicates that Europe is, at present, the commercial core of the business. The total order volume reaches 233 thousand euros, and the charts demonstrate sustained growth over time; however, this progress coexists with a very high average ticket of €4.66K, confirming that our products are high-value items and that purchases tend to be punctual rather than recurring.

Nevertheless, most accounts are classified as Occasional and are also categorized as High Risk, which means the business operates without long-term agreements and lacks mechanisms that ensure recurring purchases. The recency data reinforces this: many companies have gone more than 400 days without placing an order, revealing a structural lack of recurrence and the absence of a renewal cycle.

The company is indeed growing, but it is doing so by relying on one-off sales rather than building sustainable relationships that would provide stability and predictability in revenue streams.

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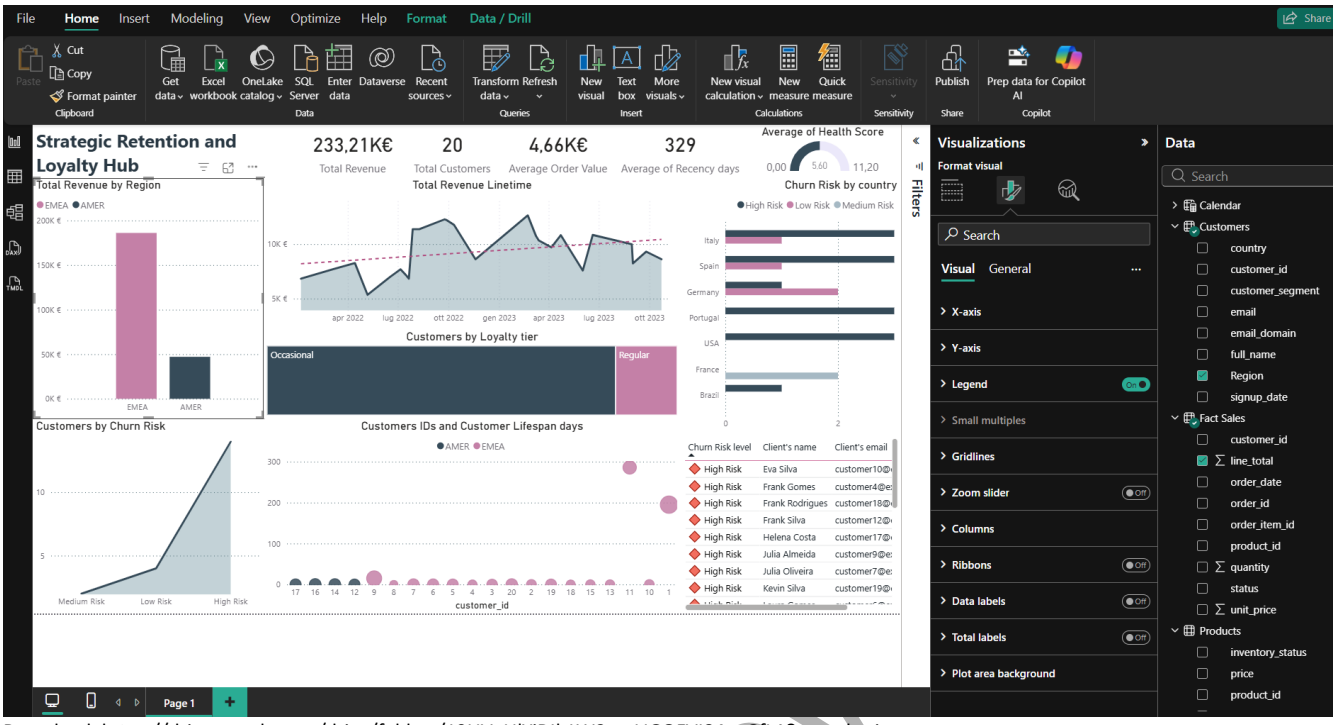

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8. File Structure (for GitHub)

README.md - (overview & instructions,how to use)

Carpet "SQL"

- week3schemaMySQL (it contains all the schema for the tables and its connections for MySQL)
- week3SchemaSQLite (it contains all the schema for the tables used for our resources and translated with Copilot)
- weekBusQueriesMySQL(it contains all the queries to resolve the business needs)
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